Kenneth Cula

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Experience

AT&T — **Atlanta, GA**
Embedded & Network Engineer — *Jul 2025 – Present*

- Developed a Linux embedded application in **C** using **Legato Application Framework** to coordinate dual Snapdragon Auto 5G modems through event-driven IPC, state synchronization, and network failover handling.
- Implemented modem state synchronization and handoff logic between **LTE** and **5G NR** interfaces to support network transition for legacy IoT platforms. Targeting deployment across automotive and large-scale IoT modem platforms.
- Built a Python **network automation pipeline** using Paramiko SSH orchestration and Pandas-based audit processing to validate configurations across 300+ enterprise routers, reducing manual audit time by 99%.
- Developed an automated **workflow orchestration system** using Selenium and React to generate and submit large-scale operational work tickets, increasing processing throughput by 500%.

MercuryVote — **New York, NY**
Full Stack Engineer — *Dec 2024 – Jul 2025*

- Designed and deployed responsive company front-end using React, TypeScript, and TailwindCSS, ensuring high performance and seamless user experience. Hosted on Vercel for optimized deployment and scalability.
- Implemented back-end support for secure authentication and authorization workflows using **OAuth2** and **JWT**.
- Implemented advanced security measures including **PostgreSQL Row-Level Security** and discretionary access controls to protect sensitive election data.

Projects

AT&T — **Atlanta, GA**
Innovation Labs Lead — *Jan 2025 – Present*

- Led and delivered hands-on technical workshops for 50+ participants across the TDP and AT&T community, covering **embedded systems**, distributed systems, soldering, and network fundamentals.
- Architected and managed development of a Financial Management System for the TDP community, leading a 15-person cross-functional team across frontend and backend from concept through production delivery.

Cornell University — **Ithaca, NY**
Distributed Systems Labs and Framework — *Jan 2024 – May 2024*

- Implemented a fault-tolerant distributed ledger using **Multi-Paxos** replication and primary-backup failover to maintain consistency under simulated server failures.
- Designed **sharding** and **load-balancing** mechanisms to improve distributed workload scalability and replica coordination efficiency.

Education

Cornell University — **Ithaca, NY**
Master of Engineering in Computer Science — *Jan 2024 - Dec 2024*
Bachelor of Arts in Computer Science — *Aug 2020 - Dec 2023*

Relevant Coursework: Operating Systems • Distributed Systems • Database Systems • Computer Networks • ML

Tech & Tools

Languages: Python, Java, C, C++, Rust, Go, TypeScript, SQL, Bash/Shell
Embedded & Systems: Embedded Linux, Legato, Multithreading, IPC, TCP/IP, CAN/LIN
Networking: 5G NR, LTE, Ethernet, Routing, SSH, Network Automation
Tools: Git, Linux, Bash, Docker, Pandas, Make, Wireshark